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Enhanced immune responses to mRNA compared with adenoviral vector COVID-19 vaccines during pregnancy: implications for pandemic preparedness

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ABSTRACT

Introduction: Multiple SARS-CoV-2 vaccine types are available but the optimal platform for use in pregnancy remains unclear. This study compared humoral and cellular immune responses in blood and breastmilk following COVID-19 vaccination with different vaccine types during pregnancy.

Methods: In a prospective cohort study, pregnant individuals (85 % naïve for COVID-19) received at least one dose of either the adenoviral vector vaccine (AVV) ChadOx1-S ($N = 17$) or a messenger RNA vaccine (mRNAV) BNT162b2 ($N = 42$) or mRNA-1273 ($N = 5$) during pregnancy. Blood samples were taken at different timepoints before and after each vaccine dose, and at delivery. Breastmilk samples were collected up to 12 weeks postpartum. Follow-up continued until 28–35 days after postpartum mRNA booster vaccination (BNT162b2: $N = 23$; mRNA-1273: $N = 25$). The primary outcome was comparison of immune responses between the AVV and mRNAV group.

Results: mRNAV recipients showed significantly higher anti-RBD IgG titres, with greater avidity and neutralizing capacity, and stronger S1- and S2-specific CD154 CD4+ T-cell responses up to day 28 after the second vaccine dose. Cord blood from mRNAV recipients had significantly higher anti-RBD IgG and NAb titres at delivery compared to AV vaccinated individuals. After postpartum mRNA booster vaccination, immune responses were comparable between both groups.

Conclusion: mRNA COVID-19 vaccination during pregnancy elicited more rapid and robust immune responses compared to AV vaccination. These findings suggest that mRNA vaccines may be preferable for emergency use during pregnancy and support their use in future maternal vaccine development and in pandemic preparedness recommendations.

Clinical trial registry: [Clinicaltrials.gov](https://clinicaltrials.gov) identifier: NCT05618548.

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1. Introduction

Pregnant individuals and their offspring are at increased risk of COVID-19 related complications [1–3]. Vaccination is therefore key to prevent adverse outcomes during pregnancy and at birth. Since early 2021, several COVID-19 vaccines, based on different platforms, have been licensed and implemented worldwide [4]. However, pregnant individuals were initially excluded from pre-licensure vaccine trials, raising concerns about vaccine safety, immunogenicity and effectiveness in this population [5–7]. Current evidence primarily focused on messenger ribonucleic acid-based COVID-19 vaccines (mRNAs) (BNT162b2 and mRNA-1273) [8]. Several studies showed that COVID-19 mRNA vaccination is safe across all trimesters of pregnancy [9], with reactogenicity profiles, as well as humoral and cellular immune responses largely comparable between pregnant and non-pregnant individuals, although data on cell-mediated immunity remain limited [8,10–12]. Besides maternal protection, COVID-19 vaccination during pregnancy does also protect the neonate. Different studies documented the presence of SARS-CoV-2-specific IgG antibodies in cord blood [13–15], as well as anti-SARS-CoV-2 IgG and IgA antibodies in breast-milk (BM) following mRNA vaccination during pregnancy [16].

In contrast, evidence on the administration of non-mRNA COVID-19 vaccines, such as adenoviral vector vaccines (AVVs), in pregnancy remains limited [10]. Most existing studies focused on the Ad26.COV2-S vaccine, with limited data on ChAdOx1-S [17–19]. Comparative studies between COVID-19 vaccine platforms in pregnant individuals are also scarce. A meta-analysis, although limited to eight studies, compared the safety and reactogenicity profiles of mRNA versus non-mRNA COVID-19 vaccines in pregnancy. This meta-analysis reported a lower incidence of fetal or neonatal death but a higher frequency of maternal fever following non-mRNA compared with mRNA vaccination during pregnancy [20]. However, this analysis did not evaluate immunogenicity or the transfer of maternal antibodies to the neonate. Immunogenicity profiles in pregnant individuals vaccinated with AVVs versus mRNAs were assessed in a study by Atyeo et al. They found that mRNA vaccination induced higher antibody titers in both maternal and cord blood, along with enhanced vaccine-induced functions and Fc receptor binding compared with vaccination with an AVV [17]. However, all AV-vaccinated participants in that study received the Ad26.Cov2-S vaccine platform, while the ChAdOx1-S platform was not included.

The present study addressed the current knowledge gap regarding reactogenicity and immunogenicity of different COVID-19 vaccine platforms in pregnant individuals. The primary outcome involved comparison of humoral and cellular immune responses in those receiving primary vaccinations during pregnancy with the AVV ChAdOx1-S versus mRNAs (BNT162b2 and mRNA-1273). Therefore, a multidimensional approach was used. Besides maternal immune responses, the neonatal impact was assessed through transplacental and BM antibody transfer. Additionally, postpartum mRNA-booster humoral and cellular immune responses were assessed in those primary vaccinated during pregnancy. These findings will contribute to identifying optimal selection of vaccine platforms for use during pregnancy and, in the context of pandemic preparedness, support the rapid development of vaccines against newly emerging viruses, suitable for administration during pregnancy.

2. Methodology

This multicentric prospective phase IV cohort study was performed in accordance with the Declaration of Helsinki, ICH-GCP, and Belgian law procedures for clinical trials. The study was approved by the ethics committee of the Antwerp University Hospital, Belgium ([Ethics approval number: 21/05/077 - Clinicaltrials.gov identifier: NCT05618548](#)).

2.1. Study population

Participants were enrolled at three Belgian sites: University of Antwerp, Antwerp (coordinating centre); CHU St Pierre, Brussels; Erasme Hospital, Brussels, between February 16 and August 12, 2021. Written informed consent was obtained from all participants after clear explanation of the study. Eligibility criteria included being pregnant and intending to receive a COVID-19 vaccine during pregnancy. Exclusion criteria were age under 18, refusal of COVID-19 vaccination during pregnancy, severe immunological disease and/or immunosuppressive treatment during pregnancy.

2.2. Study vaccines

Study vaccines were administered as part of the Belgian COVID-19 vaccination program. Participants received two doses of either the licensed AVV ChAdOx1-S (Oxford-AstraZeneca) or two doses of the same licensed mRNA: BNT162b2 (Pfizer-BioNTech) or mRNA-1273 (Moderna). Vaccines were injected intramuscularly in the left or right deltoid muscle. At least the first COVID-19 vaccine dose was given during pregnancy. In April 2021, the Belgian authorities restricted ChAdOx1-S use to individuals over 41 years due to reports of rare thrombotic events, particularly in young females. This restriction limited the AVV cohort size. Postpartum booster vaccination was performed with BNT162b2 or mRNA-1273. The interval between the first (start primary vaccination) and third vaccine dose (booster vaccination) ranged from 175 to 316 days.

2.3. Study procedures

Maternal serum and heparinized blood samples were collected at several timepoints before and after COVID-19 vaccination: pre-dose 1 (D1_DAY00), pre-dose 2 (D2_DAY00), and pre-dose 3 (D3_DAY00); at day 7, day 28, and day 159 after dose 2 (D2_DAY07; D2_DAY28; D2_DAY159); 28 to 35 days after dose 3 (D3_DAY28–35). For the group primed with ChAdOx1-S, an additional timepoint 49 to 56 days after dose 1 (D1_DAY49–56) was added to align with timepoint D2_DAY28 in the mRNA group, accounting for different recommended intervals between doses (ChAdOx1-S: approximately 12 weeks; mRNA: approximately three weeks).

To assess neonatal immunity, cord blood (deliv_cord) and maternal blood (deliv_mat) were collected at delivery. BM samples were collected at 4, 8, and 12 weeks postpartum. An overview of all study procedures is provided in Supplement 1.

2.4. Safety assessments

Because COVID-19 vaccination occurred within the national vaccination program, routine post-vaccination care was provided by external healthcare workers. Participants completed a diary to register side effects, medication, and hospitalization up to 14 days after each vaccine dose. All serious adverse events (SAEs) occurring during the complete study period were recorded. At each study visit, medical history of diseases in the household, mainly respiratory diseases, was queried. Every pre-infection or breakthrough infection (BTI) with COVID-19 (Supplement 2) was noted in the participant's file.

2.5. Laboratory

Immunogenicity was assessed through the following parameters. Detailed laboratory methods can be found in Supplement 3.

1. Immunoglobulin G (IgG) and Immunoglobulin A (IgA)

SARS-CoV-2 specific anti-receptor-binding domain (anti-RBD) IgG (BAU/mL; lower limit of detection (LLOD): 5 BAU/mL) antibody levels

were measured in serum at all timepoints using a validated Enzyme-Linked Immunosorbent Assay (ELISA).

In all collected BM samples, IgG (BAU/mL; LLOD: 0.06 BAU/mL) and IgA (ng/mL; LLOD: 0.02 ng/mL, except for anti-S2 IgA: 0.04 ng/mL) antibody levels against RBD, S1, and S2 were evaluated using a validated Multiplex ImmunoAssay.

All assays targeted the Wuhan strain.

2. Neutralizing antibodies (NABs)

NAb levels with 50 % neutralizing titre (NT50) (IU/mL; variable LLOD: Wuhan 50–89 IU/mL; Delta 64–77 IU/mL; Omicron BA.1 64–102 IU/mL; Omicron BA.5 58–75 IU/mL) were assessed, as previously described [21,22], in serum samples collected at D1_DAY00, D1_DAY49–56, D2_DAY28, D2_DAY159, D3_DAY00, D3_DAY28–35, and at delivery. Wuhan strain titres were always measured; other SARS-CoV-2 strains were tested depending on sample timing and circulating variants of concern (VOCs).

3. Avidity of anti-RBD IgG antibodies

To have an indication of the quality of antibodies, avidity of anti-RBD IgG antibodies was assessed in serum at D1_DAY49–56 and D2_DAY28. Avidity was assessed by measuring the binding dissociation rate (K_{off} (1/s)) using the BioLayer Interferometry (BLI) method. BLI measurements were performed with the ForteBio HTX Octet instrument and ForteBio AR2G biosensors.

4. T-cells

Peripheral blood mononuclear cells (PBMCs) were isolated within 4 h after collection of heparinized blood. T-cell responses were evaluated at D1_DAY49–56 and D2_DAY28 by flow cytometry and ELISpot, and at D3_DAY00 and D3_DAY28–35 by ELISpot only. For flow cytometry analysis, PBMCs were stimulated with peptide pools of SARS-CoV-2 Wuhan Spike protein (PepMix™ SarsCov2 SUB1 and PepMix™ SarsCov2 SUB2, JPT, Berlin, Germany). Cells were stained with surface antibodies to CD3 (BV711), CD4 (BV450), CD8 (PeCy7), and CD45 RA-PE and intracellular antibodies to CD154 APC-Cy7, tumor necrosis factor (TNF) α AF700, interleukin (IL) 2 APC, interferon (IFN) γ FITC, CD137 PeCy5.

For ELISpot analysis, peptide pools of SARS-CoV-2 Wuhan and variants Spike proteins (PepMix™ SARS-CoV-2 Wuhan and PepMix™ SARS-CoV-2 B.1.1.529/BA.4 & BA.5/Omicron) were used, with dividing subunit 1 (S1) and 2 (S2) of the Spike protein in different vials. Peptides are referred to in the text as WU1, WU2, O1 and O2.

2.6. Statistics

Baseline characteristics are expressed as median with interquartile range (IQR) or as absolute numbers with percentages. Only laboratory results from blood and BM samples collected within the predefined acceptable time windows were included (Supplement 4).

For antibody titres, mean estimates with 95 % confidence interval (CI) were calculated from vector generalized linear models (VGLM) using antibody titres on log scale assuming normal distribution (no censoring) or censored normal distribution (censoring occurred <60 %); median and IQR are reported if censoring occurred in ≥ 60 %. The transplacental transport ratio was calculated as the mean of the cord titres divided by maternal titres at delivery. Differences between the two vaccine groups (mRNAV, AVV) were investigated accordingly: VGLM assuming normal distribution (no censoring in both groups) or censored normal distribution (censoring occurred <60 % in both groups) reporting p -value from likelihood ratio test. In situations with more censoring, group comparison was done using Wilcoxon rank sum test, no estimate of group difference is presented. For avidity and T-cell

immunity, median and IQR are reported for each group and Wilcoxon rank sum test was used for comparison. Only T-cell responses measured from samples with >80 % cell viability were included for analysis. As a sensitivity analysis, analysis was performed excluding observations after pre- or breakthrough infection. All analyses were performed using the statistical software R (version 4.4.2) and the VGAM package (version 1.1–13) [23,24].

3. Results

3.1. Demographics

A total of 64 pregnant individuals were included in the study (Fig. 1). A summary of demographic and clinical characteristics of the participants is listed in Table 1. Both cohorts, the mRNAV and AVV group, were largely comparable in baseline characteristics. Notable differences were observed in the proportion of healthcare workers, reflecting changes in the Belgian vaccination strategy from April 2021; gestational age (GA) at first COVID-19 vaccine dose; and coverage of influenza vaccination during pregnancy.

3.2. Safety

Adverse events (AEs) were reported after vaccination with the first, second, and third COVID-19 vaccine dose in both vaccine groups. All reported AEs were mild and mostly self-limiting, resolving within seven days post-vaccination. A full overview of reported AEs can be found in Supplement 5.

No SAEs were reported in mothers during the study period. Three cases of late preterm birth occurred in the mRNAV group (range GA: 36.4–36.7 weeks): one involved twin pregnancy, while the other two had no identified cause. None of the infants required prolonged neonatal hospitalization. During the study period, two infants born to mRNA-vaccinated mothers, infected with COVID-19 infection at 22 and 39 days of age, were hospitalized out of caution. Additionally, four (mRNAV: $N = 3$; AVV: $N = 1$) infants were hospitalized for reasons unrelated to COVID-19, including respiratory tract infections and hyperbilirubinemia.

3.3. Maternal immune responses after COVID-19 vaccination

3.3.1. At baseline before vaccination (D1_DAY00)

Prior to vaccination, ten participants (mRNAV $N = 9$; AVV $N = 1$) had evidence of prior SARS-CoV-2 infection based on the study definition (Supplement 2). Among these, eight individuals (mRNAV $N = 7$; AVV $N = 1$) had detectable anti-RBD IgG antibodies at D1_DAY00. None of the pre-infected participants had detectable Wuhan NABs at baseline (Fig. 2A–B; Supplement 6–7).

3.3.2. After primary vaccination - short-term (D1_DAY49–56; D2_DAY00; D2_DAY07; D2_DAY28)

Following primary COVID-19 vaccination, an increase in anti-RBD IgG levels and Wuhan NABs was measured in both groups, with significantly higher titres in mRNAV recipients at all early timepoints after the first and second vaccine dose (Fig. 2A–B; Supplement 6–7). Among mRNA-vaccinated pregnant individuals, peak IgG antibody titres were reached earlier after primary vaccination, with the highest anti-RBD IgG antibody titre at D2_DAY07 in the mRNAV versus at D2_DAY28 in the AVV group (Fig. 2C).

For 13 participants from the AVV group and 35 from the mRNAV group, anti-SARS-CoV-2 NABs against the Delta strain were measured at D2_DAY28. The Delta NAB titres were significantly higher in the mRNAV group (ln(IU/mL): 4.40 [95 % CI 3.77–4.87] – $N = 16$ (46 %) < LLOD) compared to the AVV group (ln(IU/mL): 3.65 [IQR 3.65–3.65] – $N = 11$ (85 %) < LLOD) ($p = 0.044$).

The avidity of anti-RBD IgG antibodies at D2_DAY28 in mRNAV

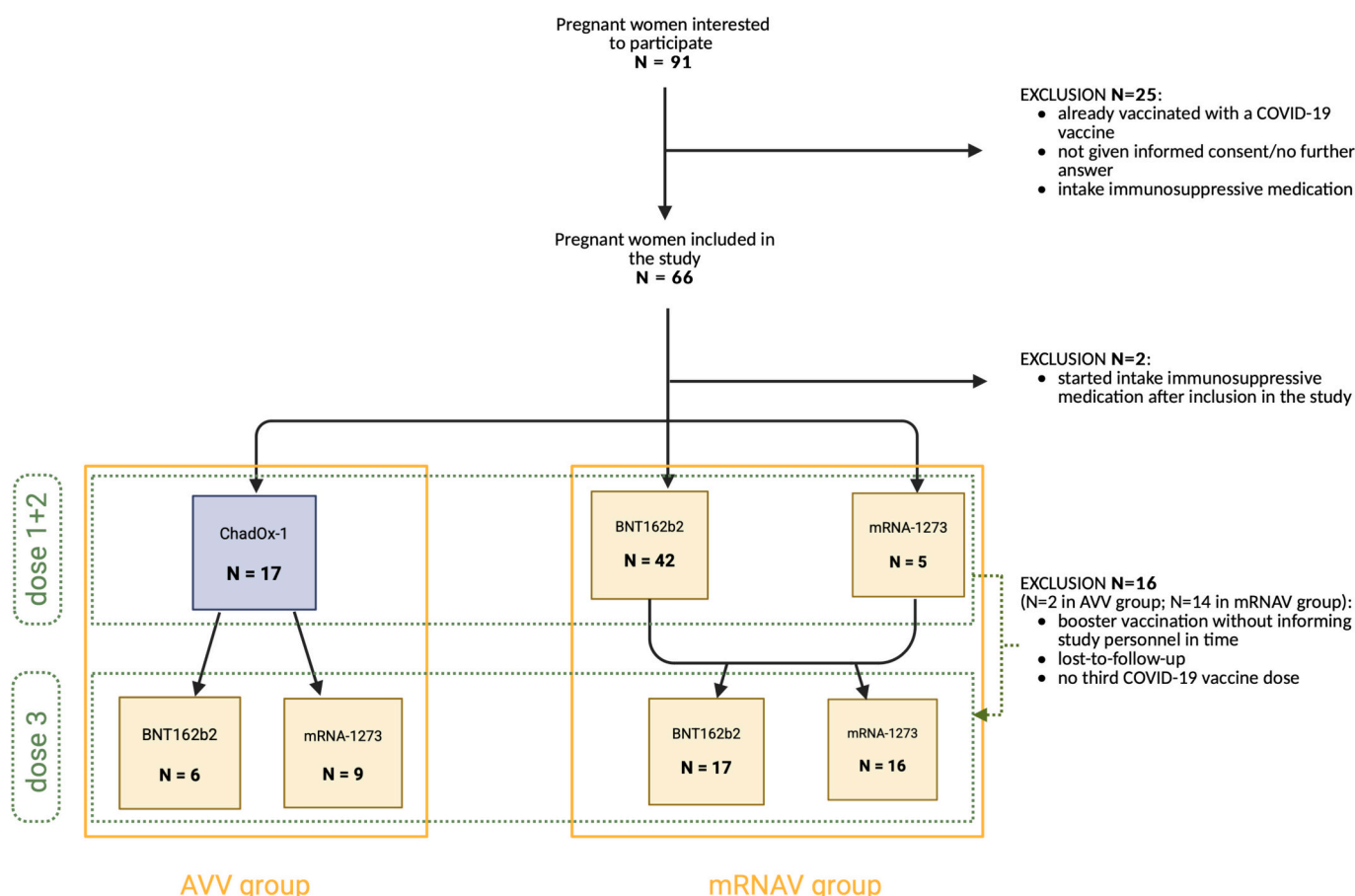


Fig. 1. Flow diagram recruitment COVID-19 vaccination study during pregnancy.

recipients ($1/K_{\text{off}}$ (s): 8108 [IQR 6381–14,864]) was significantly higher compared to avidity at D1_DAY49–56 and D2_DAY28 in the AVV group ($1/K_{\text{off}}$ (s): D1_DAY49–56: 1032 [IQR 500–1730]; D2_DAY28: 4419 [IQR 1964–9356]) ($p < 0.001$; $p = 0.017$, respectively) (Fig. 2E).

At D1_DAY49–56 (AVV group only) and D2_DAY28 (both groups), S1- and S2-specific CD4+ and CD8+ T-cell responses were induced in both study groups, with overall higher CD4+ T-cell responses. At D2_DAY28, participants vaccinated with a mRNAV had significantly higher levels of S1- and S2-specific CD154 CD4+ T-cells, and CD4+ T-cells producing IL-2. When comparing D1_DAY49–56 responses in the AVV group to D2_DAY28 responses in the mRNAV group, significantly higher S1- and S2-specific CD154 CD4+ T-cell responses, and CD4+ T-cell responses producing IFN γ and IL-2 were observed in the mRNAV group. Overall S2-specific CD8+ T-cell responses and, more precisely, S2-specific CD8+ T-cell responses secreting TNF α were significantly higher in the AVV compared to the mRNAV group at D2_DAY28 (Fig. 3; Supplement 8).

ELISpot analysis showed significantly higher WU2 IFN γ frequencies at D2_DAY28 in the mRNAV group compared to responses at D1_DAY49–56 and D2_DAY28 in the AVV group. No significant differences were found between the groups for WU1 frequencies (Supplement 9).

3.3.3. After primary vaccination – long-term (D2_DAY159)

Approximately six months after the first vaccine dose, a decline in anti-RBD IgG and NABs was found in both vaccine groups. Hence, significantly higher anti-RBD IgG antibody titres remained in the mRNAV group. By contrast, no significant difference in Wuhan NAB levels was detected between the study groups at this timepoint (Fig. 2A–B; Supplement 6–7). Anti-SARS-CoV-2 NABs against the

Omicron BA.1 and BA.5 strains were also measured for participants with Wuhan NAB NT50 titres >300 IU/mL at D2_DAY159 (AVV group $N = 3$; mRNAV group $N = 8$). As two of the three AVV participants had a titre below the LLOD, no further statistical analysis was performed on these Omicron data.

3.3.4. Booster vaccination (D3_DAY00; D3_DAY28–35)

Following postpartum mRNA booster vaccination (D3_DAY28–35), anti-RBD IgG and Wuhan NAB titres were comparable between vaccine groups (Fig. 2A–B; Supplement 6–7). Among participants with Wuhan NAB NT50 titres >300 IU/mL (D3_DAY28–35: AVV group $N = 14$; mRNAV $N = 22$), Omicron BA.1 and BA.5 Nabs were assessed. No significant differences were detected between the mRNAV and AVV group for either variant (BA.1: AVV 5.19 [95 % CI 4.68–5.67] – mRNAV 5.56 [95 % CI 4.90–6.16] ($p = 0.31$)) (BA.5: AVV 5.18 [95 % CI 4.44–5.84] – mRNAV 5.29 [95 % CI 4.67–5.87] ($p = 0.76$)).

For T-cell responses before (D3_DAY00) and after (D3_DAY28–35) booster vaccination, no significant differences were measured with ELISpot for WU1, WU2, O1, and O2 responses (Supplement 9).

3.4. Transfer of antibodies to the neonate

3.4.1. Transplacental transfer

In both study groups, anti-SARS-CoV-2 anti-RBD IgG antibodies and Wuhan NABs were detected in cord blood at delivery, with significantly higher titres in the mRNAV compared to the AVV group. In maternal serum at delivery, significantly higher anti-RBD IgG antibody titres were found in the mRNAV versus AVV group, but no significant difference was observed for the Wuhan NABs (Fig. 4A–B; Supplement 6–7). The median transplacental transfer ratio for anti-RBD IgG was similar

Table 1
Demographic characteristics of participants.

	mRNA (N = 47) <i>BNT162b2 or mRNA-1273</i>	AVV (N = 17) <i>ChadOx1-S</i>	p-value
MOTHER			
Maternal age (years)	31.9 (25.8–40.6)	30.3 (27.0–34.1)	0.08
Education			0.5
Primary school	1 (2 %)	..	
Secondary school	3 (6 %)	1 (6 %)	
Bachelor	13 (28 %)	8 (47 %)	
Master	24 (51 %)	8 (47 %)	
PhD	6 (13 %)	..	
Healthcare worker			0.02
Yes	27 (57 %)	15 (88 %)	
No	20 (43 %)	2 (12 %)	
Race			0.8
African	3 (7 %)	..	
Asian	1 (2 %)	..	
Caucasian	41 (87 %)	17 (100 %)	
Caucasian-African	1 (2 %)	..	
Latin American	1 (2 %)	..	
COVID-19 infection before vaccination			0.6
Yes	9 (17 %)	1 (6 %)	
No	38 (83 %)	16 (94 %)	
Trimester of pregnancy at first COVID-19 vaccine dose			0.001
1 (week 1–12)	5 (11 %)	1 (6 %)	
2 (week 12–27)	34 (72 %)	5 (29 %)	
3 (week 28–delivery)	8 (17 %)	11 (65 %)	
Other vaccines during pregnancy			0.015
Influenza	17 (36 %)	12 (71 %)	
Pertussis	45 (96 %)	17 (100 %)	>0.9
Comorbidities during pregnancy			
Diabetes	5 (11 %)	..	
Hypertension/(pre-) eclampsia	3 (6 %)	1 (6 %)	
Hypothyroidism	6 (13 %)	..	
DELIVERY			
GA at delivery (weeks)	40.0 (36.4–41.3)	39.6 (37.1–41.4)	>0.9
Mode of delivery			0.09
C-section	7 (15 %)	6 (35 %)	
Vaginal	40 (85 %)	11 (65 %)	
Induction			0.5
Yes	15 (32 %)	7 (41 %)	
No	31 (66 %)	10 (59 %)	
Missing	1 (2 %)	..	
Neonatal birth weight (g)	3415 (2440–4540)	3500 (2750–4280)	0.8

Demographic characteristics of participants included in the study. For maternal age, gestational age at delivery, and neonatal birth weight, the median with IQR is given. For the other parameters, the absolute numbers and percentages are shown. To define trimester of pregnancy, the definition by the National Health Society (NHS), UK was used.¹⁷ P-values were calculated using Wilcoxon rank sum test for maternal age, GA at delivery, and neonatal birth weight; Pearson's

Chi² test for other vaccines during pregnancy: pertussis and induction; Fisher's exact test for education, healthcare worker, race, blood type, COVID-19 infection before vaccination, trimester of pregnancy at first COVID-19 vaccine dose, other vaccines during pregnancy: influenza, and mode of delivery. No p-value was assessed for comorbidities during pregnancy. The p-value is considered significant if $p < 0.05$ (in bold).

between groups (mRNAV: 1.15 – AVV: 1.13). For Wuhan NABs, the median transplacental transfer ratio was 1.00 in both groups (Fig. 4C–D). A positive correlation was found between maternal and cord blood for both anti-RBD IgG and Wuhan NAb titres at delivery (Fig. 4E–F).

3.4.2. Anti-SARS-CoV-2 IgG and IgA antibodies in breastmilk

Anti-SARS-CoV-2 anti-RBD, anti-S1, and anti-S2 IgG and IgA antibodies were detected in BM at week 4, week 8, and week 12 postpartum after COVID-19 vaccination during pregnancy (Fig. 5). No significant differences in BM IgG and IgA antibodies were seen between the AVV and mRNAV group at any timepoint. An increase in BM IgG and/or IgA antibodies was present after (booster) COVID-19 vaccination and/or self-reported BTI, though this pattern was not consistent across all participants.

3.5. Sensitivity analysis

3.5.1. Humoral immunity

To assess the potential impact of SARS-CoV-2 pre-infection and BTI on immune responses upon vaccination, a sensitivity analysis was performed excluding all observations after pre-infection or BTI. Overall, humoral immunity results were consistent with the main analysis. However, three differences were noted. The previously observed significant difference in anti-RBD IgG titres in maternal serum at D3_DAY00 between groups was no longer statistically significant ($p = 0.094$). At D3_DAY28–35, Wuhan NABs were significantly higher in the AVV compared to the mRNAV group ($p = 0.033$). At 12 weeks postpartum, anti-S2 IgA antibody titres in BM were significantly higher in the AVV versus mRNAV group ($p = 0.045$).

3.5.2. Cellular immunity

Following exclusion of data after pre-infection and BTI, some differences emerged when comparing cellular immune responses with the main analysis: the difference of S1- and S2-specific CD4⁺ T-cell responses producing IL-2 at D2_DAY28 between both vaccine groups was not significant anymore (S1: $p = 0.059$; S2: $p = 0.079$). When comparing D1_DAY49–56 results in the AVV group with D2_DAY28 results in the mRNAV group, S1-specific CD8⁺ T-cell responses producing TNF α became significantly higher in the AVV versus mRNAV group ($p = 0.038$).

For WU2 IFN γ T-cell responses at D2_DAY28, no significant difference was measured anymore between the mRNAV and AVV group ($p = 0.114$).

4. Discussion

This prospective cohort study reports comparative data on maternal humoral and cellular immune responses following COVID-19 vaccination in pregnancy, using two different vaccine platforms: mRNAVs (BNT162b2 or mRNA-1273) and AVV (ChadOx1-S). A key strength of the study is the inclusion of a unique cohort of pregnant individuals primarily vaccinated with ChadOx1-S, a group underrepresented in prior research. Importantly, no SAEs were reported among vaccinated mothers, reaffirming that COVID-19 vaccination during pregnancy is safe [9]. Although two infants born to mRNA-vaccinated mothers were hospitalized with COVID-19, both admissions were precautionary due to fever in infants below three months of age.

After primary vaccination, significantly higher anti-RBD IgG titres

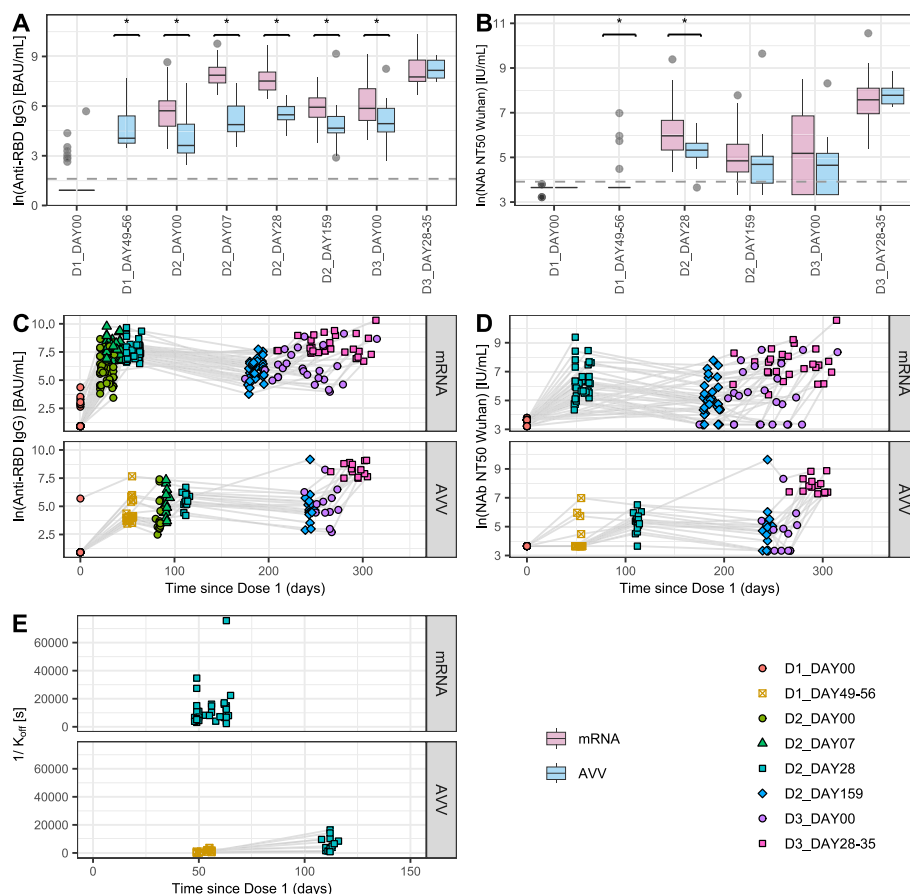


Fig. 2. Humoral immunity in **maternal blood** after COVID-19 vaccination with different vaccine platforms during pregnancy. (A & B) Boxplot representing the anti-SARS-CoV-2 anti-RBD IgG titres (A) and Wuhan NT50 NAb titres (B) in serum at different timepoints before and after AV vs mRNA COVID-19 vaccination. Ln values are used and values below the LLOD (indicated by the dashed line) are replaced by LLOD/2. Significant differences ($p < 0.05$) are indicated with “*”. For all participants, the third vaccine dose was a mRNAV administered in the postpartum. The results at D1_DAY49–56 in the AVV group were compared with the results at D2_DAY28 in the mRNAV group. (C & D) Spaghetti plot representing the anti-SARS-CoV-2 anti-RBD IgG titres (A) and Wuhan NT50 NAb titres (B) in serum through time before and after AV vs mRNA COVID-19 vaccination. Ln values were used and values below the LLOD are replaced by LLOD/2. The interval between the first (start primary vaccination) and third vaccine dose (booster vaccination) ranged from 175 to 316 days. This interval difference had no significant impact on immune responses. (E) Graph representing the avidity of the anti-RBD IgG antibodies by using $1/K_{off}$ (K_{off} = dissociation rate) measurements. K_{off} is used instead of K_{on} (K_{on} = association rate), because K_{off} is concentration independent. The lower the K_{off} , the better the avidity.

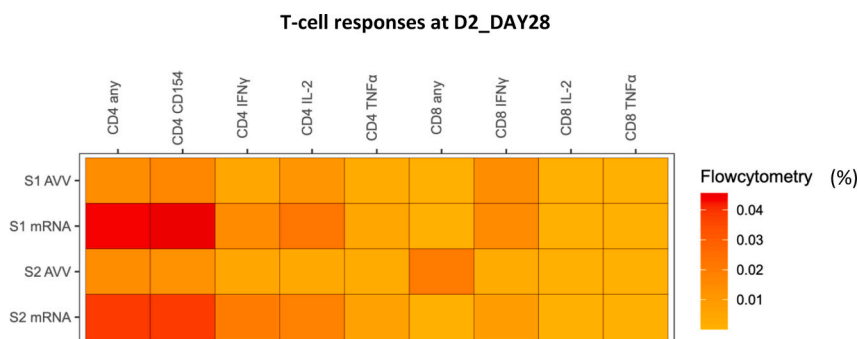


Fig. 3. Heatmap representing the median S1- and S2-specific CD4+ and CD8+ T-cell responses measured via flow cytometry at day 28 after COVID-19 vaccine dose 2, for both the mRNAV and AVV group. Each line represents S1- or S2-specific responses for a vaccine group. Each column represents a CD4+ or CD8+ specific T-cell response. Each box represents the median response of all participants at D2_DAY28.

with a greater avidity, higher Wuhan NAb (NT50) titres, and enhanced CD4+ T-cell responses and WU2 IFN γ frequencies were observed in recipients of mRNAVs compared to the AVV group. Following postpartum mRNA booster vaccination, immune response differences between the groups diminished, with no significant difference in antibody titres and T-cell responses between the groups one month after booster

vaccination.

At delivery, significantly higher titres of anti-RBD IgG and Wuhan NABs were detected in cord blood from mRNA- compared to AV-vaccinated mothers, with a positive correlation between maternal and cord blood antibody titres in both groups. This confirms efficient transplacental antibody transfer and highlights the potential for

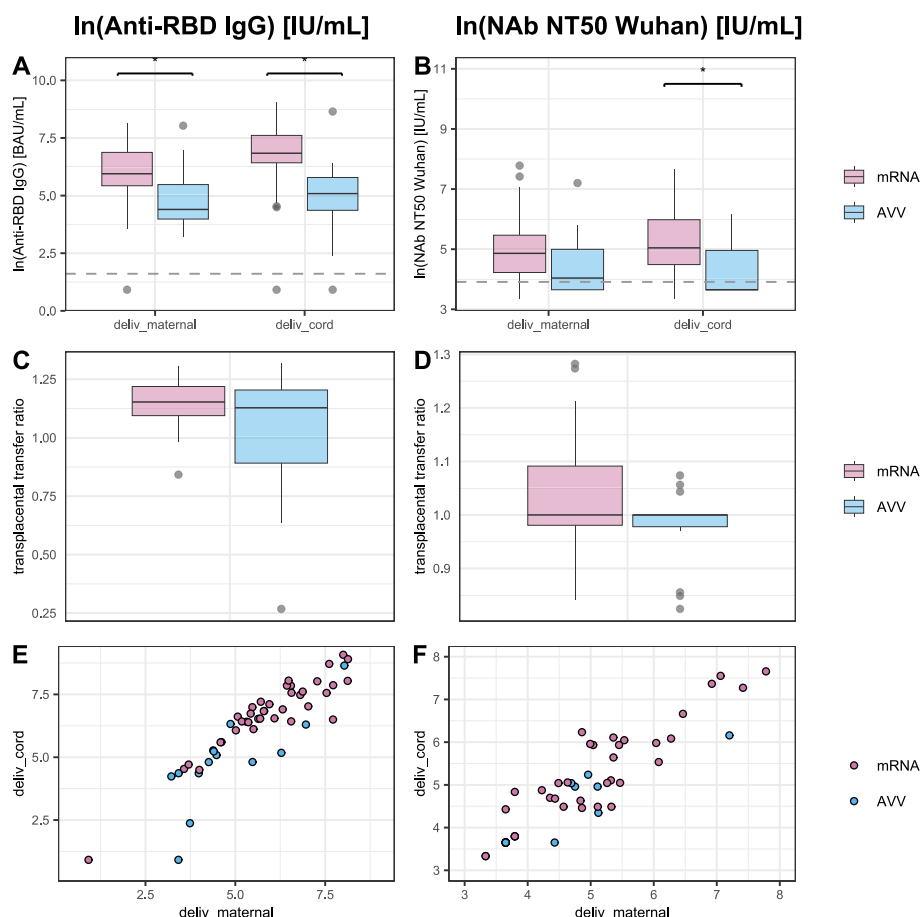


Fig. 4. Maternal and cord blood anti-SARS-CoV-2 anti-RBD IgG and Wuhan NT50 NAb titres at delivery after COVID-19 vaccination with different vaccine platforms during pregnancy; transplacental transfer ratio; and cord and maternal antibody correlation. (A&B) Boxplot representing the anti-SARS-CoV-2 anti-RBD IgG titres (A) and Wuhan NT50 NAb titres (B) in cord and maternal blood at delivery after COVID-19 vaccination during pregnancy. Ln values are used and values below the LLOD (indicated by the dashed line) are replaced by LLOD/2. Significant differences ($p < 0.05$) are indicated with “*”. Six (35 %) women in the AVV group and 46 (98 %) women in the mRNA group completed primary vaccination before delivery. The other participants ($N = 11$ (65 %) in the AVV group; $N = 1$ (2 %) in the mRNA group) got one vaccine dose during pregnancy. (C) Boxplot showing the transplacental transfer ratio of anti-SARS-CoV-2 anti-RBD IgG at delivery after COVID-19 vaccination during pregnancy. The median transplacental transfer ratio of anti-RBD IgG is 1.15 (Q1: 1.09, Q3: 1.22) in the mRNA group and 1.13 (Q1: 0.88, Q3: 1.21) in the AVV group. (D) Boxplot showing the transplacental transfer ratio of anti-SARS-CoV-2 Wuhan NT50 Nabs at delivery after COVID-19 vaccination during pregnancy. The median transplacental transfer ratio of Wuhan Nabs is 1.00 (Q1: 0.98, Q3: 1.09) in the mRNA group and 1.00 (Q1: 0.97, Q3: 1.00) in the AVV group. (E&F) Scatterplot representing the correlation between maternal and cord blood anti-SARS-CoV-2 anti-RBD IgG titres (E) and Wuhan NT50 NAb titres (F) at delivery after COVID-19 vaccination during pregnancy. Ln values are used and values below the LLOD are replaced by LLOD/2. Six (35 %) women in the AVV group and 46 (98 %) women in the mRNA group completed primary vaccination before delivery. The other participants ($N = 11$ (65 %) in the AVV group; $N = 1$ (2 %) in the mRNA group) got one vaccine dose during pregnancy.

maternal vaccination to confer passive neonatal protection. Furthermore, in both the mRNA and AVV group, anti-RBD IgG/IgA, anti-S1 IgG/IgA, and anti-S2 IgG/IgA antibodies were found in breastmilk, with no significant differences between vaccine platforms.

These findings are consistent with studies in non-pregnant adults [25], which have also demonstrated higher immune responses after mRNA compared to AV vaccination. Since mRNA vaccination elicits a faster and more robust humoral immune response, this platform may provide earlier and enhanced protection for pregnant individuals, especially valuable in the context of a rapidly evolving pandemic. Besides maternal protection, the transplacental transfer of SARS-CoV-2 specific antibodies provides neonatal protection. Previous studies showed that mRNA COVID-19 vaccination during pregnancy may reduce the risk of infant hospitalization due to COVID-19, especially in infants under three months of age [26,27]. As this study and former research detected antibodies in cord blood after AV vaccination during pregnancy [17], this vaccine platform potentially confers prevention against severe COVID-19 complications in neonates. In addition to transplacental transfer, COVID-19 specific IgG and IgA antibodies can

also be transmitted via BM conferring further postnatal protection. Although prior research found that breastfeeding reduces the risk of respiratory infectious diseases [28], the extent to which SARS-CoV-2 specific BM antibodies contribute to the prevention of severe COVID-19 in infants remains unclear.

As no established correlate of protection exists for COVID-19, it remains unclear whether the observed differences in immunogenicity between vaccine platforms in pregnancy translate into clinically meaningful differences in protection against severe COVID-19 in mothers and infants. However, neutralizing antibodies are widely regarded as a key component of protection against severe SARS-CoV-2 disease [29]. The earlier and higher peak NAb titres elicited by mRNA vaccination therefore support a platform preference for pregnant individuals, especially when rapid and robust protection is required in the context of a pandemic. Given the rapid emergence of new VOCs during the COVID-19 pandemic, our neutralization assays included multiple variants. Similar to the results observed for the Wuhan strain, significantly higher NAb titers against the Delta variant were detected in mRNA- compared to AV-vaccinated pregnant individuals at day 28 after dose 2. This

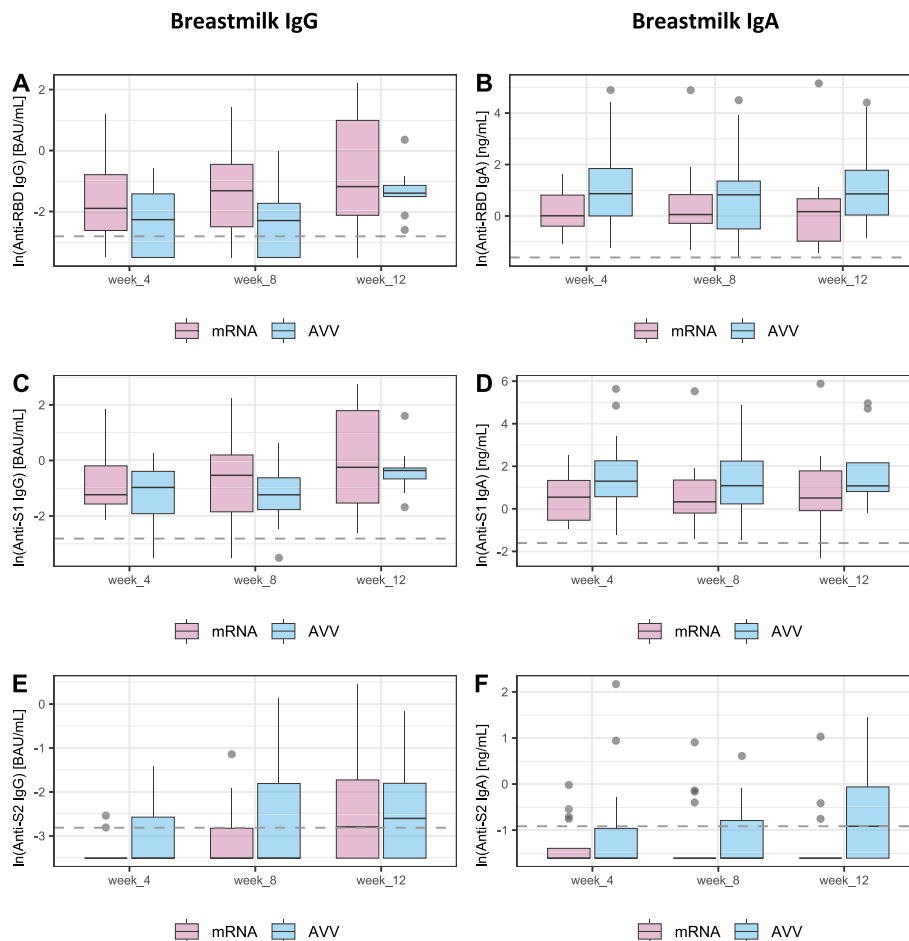


Fig. 5. Anti-SARS-CoV-2 IgG and IgA antibody titres in **breastmilk** after COVID-19 vaccination with different vaccine platforms during pregnancy. **Left column:** Boxplots representing the anti-SARS-CoV-2 anti-RBD (A), anti-S1 (C), and anti-S2 (E) IgG titres in breastmilk week 4, week 8, and week 12 postpartum after COVID-19 vaccination during pregnancy. Ln values are used and values below the LLOD (indicated by the dashed line) are replaced by LLOD/2. Six (35 %) women in the AVV group and 46 (98 %) women in the mRNA group completed primary vaccination before delivery. The other participants (N = 11 (65 %) in the AVV group; N = 1 (2 %) in the mRNA group) got one vaccine dose during pregnancy. Four (24 %) participants in the AVV group got their second vaccine dose between delivery and week 4 postpartum; four (24 %) between week 4 and week 8 postpartum; and three (18 %) between week 8 and week 12 postpartum. The woman not fully primed before delivery in the mRNA group got her second vaccine dose between week 4 and week 8 postpartum. **Right column:** Boxplots representing the anti-SARS-CoV-2 anti-RBD (A), anti-S1 (C), and anti-S2 (E) IgA titres in breastmilk week 4, week 8, and week 12 postpartum after COVID-19 vaccination during pregnancy. Ln values are used and values below the LLOD (indicated by the dashed line) are replaced by LLOD/2. Six (35 %) women in the AVV group and 46 (98 %) women in the mRNA group completed primary vaccination before delivery. The other participants (N = 11 (65 %) in the AVV group; N = 1 (2 %) in the mRNA group) got one vaccine dose during pregnancy. Four (24 %) participants in the AVV group got their second vaccine dose between delivery and week 4 postpartum; four (24 %) between week 4 and week 8 postpartum; and three (18 %) between week 8 and week 12 postpartum. The woman not fully primed before delivery in the mRNA group got her second vaccine dose between week 4 and week 8 postpartum.

suggests that the platform advantage of mRNAs likely extends to other VOCs with comparable antigenic characteristics. Nevertheless, the Omicron variant has shown substantially greater immune escape [30,31], so extrapolation of these findings to Omicron and future variants should be made with caution.

Overall, CD4⁺ T-cell responses were more prominent than CD8⁺ T-cell responses after vaccination in both groups, consistent with previous findings [32]. As virus-specific CD4⁺ T-cells are related with longer-lasting humoral immunity [33], the stronger CD4⁺ T-cell responses observed in the mRNA group could contribute to more durable protection.

Given that COVID-19 hybrid immunity provides better protection than vaccination or natural infection alone [34], a sensitivity analysis excluding all data following pre- or breakthrough infection was performed to confirm the robustness of the findings. Most results remained consistent with the main analysis, however, some differences emerged. After postpartum mRNA-booster, significantly higher Wuhan NABs were measured in the AVV compared to mRNA group, consistent with prior

findings that heterologous vaccination strategies induce more robust humoral immunity than homologous strategies [35]. Furthermore, significantly higher anti-S2 IgA titres in BM at week 12 postpartum were found in the AVV versus mRNA group. Since BM samples were collected at fixed timepoints after delivery unrelated to timing of COVID-19 vaccination, vaccination in between two timepoints of BM collection could cause an unexpected increase in antibodies. Several participants in the AVV group received their second vaccine dose between week 4 and week 8 postpartum, possibly explaining an increase of antibodies in samples collected subsequently. Lastly, for cellular immunity, only S1- and S2-specific CD154 CD4⁺ T-cell responses remained significantly higher in the mRNA group at D2_DAY28. Overall S2-specific CD8⁺ T-cell responses and, more precisely, S2-specific CD8⁺ T-cell responses secreting TNF α were significantly higher at D2_DAY28 in the AVV group, although the small number of participants with detectable responses (AVV N = 6; mRNA N = 5) limits the reliability of this observation. While CD8⁺ T-cell responses have been suggested to play an important role in SARS-CoV-2 control [36], further studies with

larger sample size are needed to assess platform-specific differences in cellular immunity.

Vaccination during pregnancy has emerged as a critical public health strategy to protect the mother, the neonate or both from vaccine-preventable infectious diseases. The COVID-19 pandemic and its according new vaccines underscored the importance of high-quality, timely vaccine trials in pregnant individuals to provide evidence-based recommendations and to strengthen vaccine confidence and uptake during pregnancy. Also, research comparing different vaccine platforms in pregnancy is essential to guide future pandemic preparedness guidelines concerning maternal vaccine development for emergency use. The World Health Organization recently published a roadmap to support research on maternal and perinatal health in the context of infectious disease outbreaks [37], stating that data on experiences of pregnant individuals and their offspring during past infectious disease outbreaks, are vital for the development of more effective preventive health strategies for future epidemics.

Our findings suggest that platform selection may influence immunogenicity outcomes in pregnancy and should be a key consideration in future maternal vaccine development. Besides immunogenicity, logistical factors are increasingly important in vaccine platform selection. mRNAs not only elicit stronger immune responses but can also be developed within markedly shorter timelines, positioning the mRNA platform as a strong candidate to achieve CEPI's 100 Days Mission, which aims to develop a new vaccine against the next "Disease X" within 100 days [40]. However, the reliance of current mRNAs on ultra-cold storage presents challenges, especially in low- and middle-income countries. Moreover, despite the withdrawal of the ChadOx1 COVID-19 vaccine from the market, the underlying AVV platform is still used in the development of vaccines targeting Zika and Chikungunya virus [38,39]. This underscores that AVVs may still have a role beyond the acute phase of a pandemic, especially when cold-chain infrastructure or cost constraints limit mRNA deployment.

A broader comparison of vaccine platforms, incorporating both immunological performance and practical implementation is needed to inform global maternal immunization strategies and policy guidance in the context of future pandemics.

5. Conclusion

This study suggests that COVID-19 mRNAs may be preferred over AVVs for use during pregnancy, as mRNA vaccination elicited more rapid humoral immune responses, with higher antibody titres, superior neutralizing capacity and greater avidity. In addition, higher frequencies of CD4+ T-cell responses were observed following mRNA vaccination, although cellular immune responses upon different vaccine types used during pregnancy should be further investigated. Maternal antibodies induced by COVID-19 vaccination in pregnancy were effectively transported to the fetus and infant across the placenta and via breastfeeding, possibly providing protection to fetus and infant in the first months of life. These findings underscore the immunological advantages of mRNA platforms in pregnancy and have important implications for the development and selection of vaccine platforms in future pandemic maternal immunization strategies.

Strengths & limitations

- To our knowledge, this is the first COVID-19 vaccine trial comparing immunogenicity between pregnant individuals who received primary vaccination with ChadOx1-S and those who received a mRNA (BNT162b2 or mRNA-1273). The inclusion of a unique cohort of primary ChAdOx1-S vaccinated pregnant individuals addresses a significant gap in the existing literature.
- All included participants received their first-ever COVID-19 vaccine dose within the study. The majority (85 %) were SARS-CoV-2-naïve at baseline, offering a rare opportunity to assess primary vaccine

responses in this population, a scenario that is increasingly difficult in the current context of widespread prior infection and vaccination.

- Vaccine allocation was non-randomized and potential confounders were not included within the statistical analysis.
- The study was influenced by Belgian national vaccination policy, which limited the use of ChadOx1-S to individuals over 41 years of age during the study period. These policy changes resulted in unequal sample sizes between the AVV and mRNA group.
- Due to the recommended 12-week interval required between two doses of ChadOx1-S, 64 % of AVV participants received only the first vaccine dose during pregnancy. Also, most participants in the AVV group were vaccinated in their third trimester of pregnancy whereas in the mRNA group, most participants received their first COVID-19 vaccine dose in the second trimester. These differences in vaccination timing and number of doses received during pregnancy likely contributed to the lower cord blood antibody titres observed in the AVV group, consistent with the positive correlation between maternal and cord blood antibody titres.
- As the study population underrepresents certain racial groups, the conclusions should be interpreted with caution and cannot be fully generalized to the overall pregnant population.

CRediT authorship contribution statement

Larissa De Brabandere: Writing – original draft, Visualization, Validation, Investigation, Data curation. **Sereina A. Herzog:** Writing – review & editing, Visualization, Validation, Formal analysis. **Isabelle Desomere:** Writing – review & editing, Investigation. **Kevin K. Ariën:** Writing – review & editing, Investigation. **Véronique Olislagers:** Writing – review & editing, Investigation. **Daphnée Georges:** Investigation. **Nicolas Dauby:** Writing – review & editing, Investigation. **An Vercoutere:** Writing – review & editing, Investigation. **Maria Goossens:** Funding acquisition. **Pieter Pannus:** Writing – review & editing, Project administration. **Elke Leuridan:** Methodology, Conceptualization. **Arnaud Marchant:** Writing – review & editing, Conceptualization. **Kirsten Maertens:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT, OpenAI (<https://chat.openai.com/chat>) in order to improve readability and language of the work. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2025.128153>.

Data availability

After publication of all results of this COVID-19 vaccine trial during pregnancy, all deidentified participant data and other documents of interest will be made available for others upon request. Requests for data sharing can be directed to LDB.

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